



STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

Lab Teacher: Dr. Aymen TRIGUI

Course Code: 462 CCS-3

Student ID: 445801958

Section Number: 904/907

S. No

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LAB ACTIVITY - 8

(1.5 Marks)

This activity aims to use the Decision Tree algorithm for “[pima-indians-diabetes](#)” dataset.

Instructions:

Create a Decision Tree, train it to fit 75% of the data, and test your model with the remaining 25%.

The following tasks can be performed:

- Import necessary tools
- Data exploration and preparation
- Creating and training the model
- The hyperparameter splitting criteria for categorical outputs are: “gini” or “entropy.” Entropy corresponds to the ID3 algorithm. The gini is the one used in the cart algorithm. Nevertheless, it is interesting to experiment with both splitting criteria.
- The hyperparameter “max_depth” can be varied to handle the overfitting. It fixes the depth of the tree.
- The hyperparameter “min_samples_split” also can be varied; it corresponds to the minimum number of samples under which the node can be splitted.
- Tree visualization (you can visualize the tree for different hyperparameters)
- Testing and measuring performance

Dataset shape: (768, 9)

Test accuracy by tree configuration:

criterion=gini, max_depth=3, accuracy=0.7135

criterion=gini, max_depth=5, accuracy=0.7917

criterion=gini, max_depth=None, accuracy=0.7448

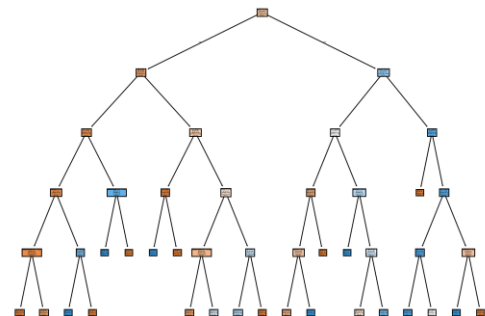
criterion=entropy, max_depth=3, accuracy=0.7135

criterion=entropy, max_depth=5, accuracy=0.7708

criterion=entropy, max_depth=None, accuracy=0.7552

Best configuration:

criterion=gini, max_depth=5, accuracy=0.7917



Marks Obtained	/1.5
Teacher's Signature:	